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Subject: Application for the Internship "Kinetic model discrimination for olefin and alcohol formation in Fischer-Tropsch synthesis" (R12-2) — IFPEN Lyon

Dear Hiring Team,

I am a final-year Industrial Chemical Engineering student at INSAT (Université de Carthage, Tunisia), specializing in Process and Environment, and I am writing to apply for the internship "Kinetic model discrimination for olefin and alcohol formation in Fischer-Tropsch synthesis" (ref. R12-2) within the Direction Conception Modélisation Procédés at IFPEN Lyon. I am available from 1 February 2027, for 6 months (3 to 9 months possible), within your January–May 2027 window.

My coursework in thermodynamics and chemical catalysis has been put into practice in an Aspen HYSYS V14 project on ethylene oxide production, where I modeled a fixed-bed tubular gas-solid catalytic reactor from its operating parameters and geometry with Langmuir-Hinshelwood kinetics of two reactions, then optimized the operating conditions by sensitivity analysis to reach 96% recovery and 97.14% purity. During my internship at SEGULA Technologies, I fitted the diffusion coefficients of a Li-ion cell model against experimental discharge data, reducing the RMSE from 202.13 mV to 40.5 mV. Implementing kinetic schemes in a reactor model and estimating their parameters against data is the same kind of quantitative work this project calls for.

IFPEN's use of a pilot-scale dataset to compare olefin and oxygenate formation mechanisms is exactly the kind of model-discrimination question I want to build my final-year project around, as it links catalysis to the design of e-fuel processes. Fischer-Tropsch chemistry and Fortran are new to me specifically, but I have programmed numerical models in MATLAB, Python and Mathematica, and my projects show that I adapt quickly to new tools and work independently.

I am drawn to this internship because it sits precisely at the intersection I want to specialize in: reaction kinetics and process modelling for sustainable fuels.

I would welcome the opportunity to discuss my application further and am available for an interview at your convenience. Thank you for your consideration.

Sincerely,
Nermine Moumen